

VINEELA MANGU

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SUMMARY

Data Analyst with 3 years of experience delivering actionable insights through cloud-based analytics, automated pipelines, and interactive dashboards.

Strong ability in SQL, Python, R, Snowflake, and BI tools (Power BI, Tableau, Excel) across healthcare, insurance, and supply chain domains.

Hands-on experience with AWS & Azure, near real-time data processing, and statistical analysis (A/B testing, time series). Practical exposure to Machine Learning and GenAI including RAG, NLP, and LLM-based applications.

SKILLS

Databases:	MySQL, Oracle, MongoDB, NoSQL, Snowflake.
Programming & Scripting:	Python, R, SQL, C, SAS.
Libraries: Keras.	Scikit-learn, Scipy, Pyspark. Pandas, numpy, Matplotlib, Pytorch, TensorFlow,
BI & Data Engineering Tools:	Tableau (Desktop, Prep), Looker, Power BI(DAX), Excel(Power Query, Power Pivot, XLOOKUP, INDEX-MATCH, Pivot Tables, VBA), Minitab, SSRS, SSIS, Data Bricks.
Cloud Platforms:	AWS (Athena, EMR, ECS, Sagemaker), Azure, GCP.
Data Analysis & techniques: Analysis.	Hypothesis testing, A/B testing, Web Scrapping, Text Analysis, Time Series
Machine Learning: Gen AI	Predictive Modeling, Supervised, Unsupervised, Agentic RAG, NLP, LLMs,
Methodologies & versions:	Agile, Slack, JIRA, Git HUB, Docker, Autodesk, GIS, visual studio code

PROFESSIONAL EXPERIENCE

Data Analytics Intern (Manufacturing & Services Domains) Margam360 Internship Period: 2020 – 2021

Supported data analysis initiatives by extracting, cleaning, and validating datasets from **MySQL, Oracle, and MongoDB** databases to help senior analysts in operational reporting and insights.

Performed **exploratory data analysis (EDA)** using **Python (Pandas, NumPy)** and **Excel Pivot Tables** to identify trends, anomalies, and data quality issues across transactional datasets.

Built automated **Excel dashboards** using **Power Query, Power Pivot, INDEX-MATCH, and VBA**, reducing manual reporting efforts by ~25%.

Assisted in developing **SQL queries** for daily, weekly, and monthly performance reports, ensuring data accuracy and consistency.

Supported ETL processes by validating data loads in **SSIS** and documenting transformation logic for downstream BI consumption.

Created basic **Tableau and Power BI reports** to visualize KPIs, trends, and variance analysis for internal reviews.

Conducted **hypothesis testing and basic statistical analysis** using **R and Mini-tab** to support business improvement initiatives.

Collaborated in **Agile teams**, took part in sprint reviews, supported task tracking in **JIRA**, and followed version control practices using **GitHub**. Documented data definitions, business rules, and reporting logic to support knowledge transfer and audit readiness.

Data Analyst Infosys Ltd Corporate Experience Period: 2021 – 2023

Worked on enterprise-scale analytics projects for insurance and retail clients, managing structured and semi-structured data across **Oracle, MySQL, Snowflake, and NoSQL** systems.

Designed and optimized **SQL-based data models** to support reporting and analytics, improving query performance by up to 35% through indexing and partitioning strategies.

Developed and automated **ETL pipelines using SSIS, Databricks, and Python**, reducing manual data processing efforts and improving data refresh reliability.

Performed advanced **EDA and statistical analysis** using **Python, R, and SAS**, uncovering operational inefficiencies and actionable insights for business stakeholders.

Built interactive **Power BI (DAX) and Tableau dashboards** integrating data from multiple sources, enabling leadership to track KPIs such as revenue trends, customer behavior, and operational SLAs.

Implemented **A/B testing and hypothesis testing** frameworks to measure the impact of process changes and digital initiatives. Leveraged **AWS (Athena, EMR)** and **Azure services** to query large datasets, ensuring scalability and cost efficiency. Supported **machine learning initiatives** by preparing feature datasets and validating model outputs for **predictive analytics** and **customer segmentation** use cases. Collaborated cross-functionally with business analysts, data engineers, and QA teams to ensure data accuracy, governance, and compliance. Followed **Agile delivery methodology**, actively participated in sprint planning, stand-ups, and retrospectives using **JIRA and Slack**. Maintained version-controlled code repositories using **GitHub**, ensuring traceability and deployment consistency.

PROJECTS

Near Real-Time Social Media Analytics Using Snowflake & Azure

Business Overview

Organizations rely on social media engagement to understand brand popularity, customer sentiment, and campaign effectiveness. Traditional batch analytics introduces delays, preventing timely decision-making. This project enables near real-time analytics on social media data to identify trending content and engagement patterns as they emerge.

Dataset Description

Twitter API data including tweets, likes, retweets, replies, timestamps, and hashtags. Near real-time ingestion using Python scripts executed on a scheduled basis. **Tech Stack**

Language: Python, SQL

Cloud Services: Azure Blob Storage, Azure Queue Storage, Azure Functions

Data Platform: Snowflake (Snowpipe, Streams, Tasks)

Visualization: Snowflake Native Dashboards **Architecture & Approach**

Developed Python scripts to fetch Twitter data and metadata using Twitter APIs.

Stored raw JSON feeds in Azure Blob Storage, triggering ingestion events.

Configured Snowpipe with Azure IAM integration for automated, event-driven data loading into Snowflake.

Implemented Snowflake Streams and Tasks for incremental processing and near real-time updates.

Built Snowflake dashboards refreshed every 30 minutes to display: Most liked

and retweeted posts

Engagement trends over time

Sentiment distribution across feeds

Outcome & Impact

Reduced data latency from hours to under 30 minutes.

Enabled early identification of viral content and engagement spikes. Demonstrated scalable near real-time analytics architecture suitable for production workloads.

Intelligent RAG-Based Healthcare Assistant (GenAI Project)

Business Overview

Healthcare professionals require quick access to accurate, context-aware medical insights. This project implements a **Retrieval Augmented Generation (RAG)** system to deliver reliable medical responses based on structured and unstructured data.

Dataset Description

Simulated doctor-patient conversations (audio & text)

Medical knowledge documents for retrieval **Tech Stack**

LLMs: Azure OpenAI (GPT models)

Vector DB: Pinecone

Speech-to-Text: OpenAI Whisper **Frontend:**

Streamlit **Frameworks:**

LangChain

Approach

Converted 60+ hours of audio conversations into text using Whisper.

Chunked and embedded medical documents into Pinecone.

Built a RAG pipeline combining semantic retrieval with LLM responses.

Developed a Streamlit web app for symptom checking and insights. **Outcome & Impact** Delivered accurate, context-aware responses for medical queries.

Reduced hallucinations compared to standalone LLMs. Demonstrated production-ready GenAI architecture with real deployment.

CERTIFICATIONS:

SQL for Data Analytics and Business Intelligence,
Tableau Desktop Certified Python for Data
Science and Analytics.

EDUCATION

Auburn University at Montgomery Jan 2023 - Dec 2024
Master of Management of Information Systems (MIS) Montgomery, AL

SR UNIVERSITY AUG 2017 - MAY 2021 Bachelor of
Technology India